

Recap

Hierarchical organization of the Name Servers used to store the DNS database

Root Name Servers are at the top of the hierarchy

Multiple instances of each Root Name Server

Top Level Domains are the children of the Root Servers

Two categories: generic and country specific

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More on TLDs

Subdomains might be associated with TLDs

ac.uk, co.in, co.uk

More [examples](#) ...

TLDs subdomains might have children (and grandchildren)

imperial.ac.uk, bankofindia.co.in, bbc.co.uk

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Second level nodes

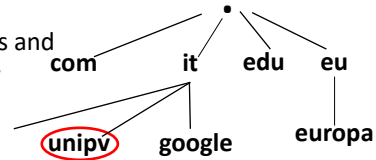
These nodes are the children of the TLD nodes and refer to the second level of the DNS hierarchy

These nodes are managed by companies, organizations or individuals responsible of the domain (or by providers offering DNS services)

They are “authoritative” for a given zone

Examples:

unipv.it, bbc.com, ucsd.edu

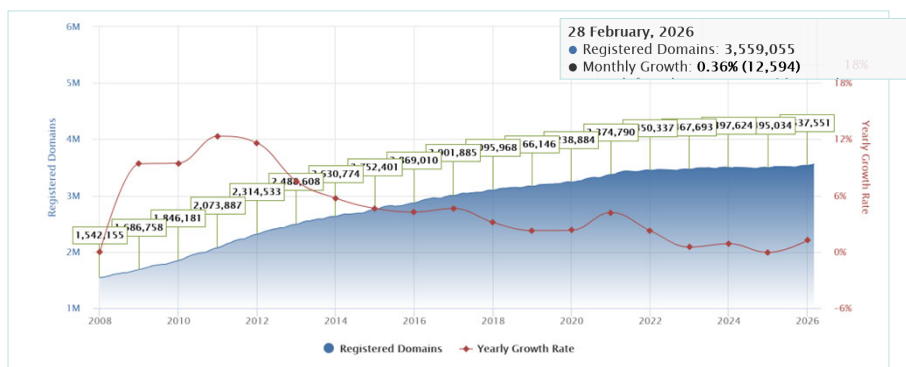


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Second level domains under it



[Source: <https://stats.nic.it/domain/registration>]

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Domain squatting: typo, combo, TLD, homograph

Microsoft vs. MikeRoweSoft
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MikeRoweSoft settles for an Xbox
Canadian student Mike Rowe, who Microsoft put under legal pressure over his Web site's name, has agreed to bury the hatchet out of court for an Xbox and some training.

Cybersquatting: Attackers Mimicking Domains of Major Brands Including Facebook, Apple, Amazon and Netflix to Scam Consumers

Domain Squatting and Brand Hijacking: A Silent Threat to Digital Enterprises
Domain squatting can lead you to malicious websites, and it might be too late to realize what actually happened.
By WAGAS NOVEMBER 15, 2023 · 6 MINUTE READ

Cybercriminals exploit brand domains as cybersquatting hits record levels
By Dr. Tim Sander Published March 17, 2026

Domain Typosquatting: Online trademark protection in the name of typing errors
NEWSROOM NOVEMBER 14, 2023

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Management

Each node of the hierarchy is identified by a name (label) and operated under the responsibility of a company or an organization (i.e., authority of the node)

Name Servers are associated with each node of the DNS hierarchy

Name Servers store a subset of all possible mappings between hostnames and IP addresses (i.e., no single repository)

Numerous geographically distributed Name Servers (and instances)

Operators of individual nodes must ensure

- 24/7 availability
- Good security levels
- Good performance levels

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Name Server diversity

At least two Name Servers per domain/node (which must not resolve to the same IP address)

- Network diversity
- Geographical diversity
- Institutional diversity
- Software and hardware diversity

Name Servers associated with the domain **unipv.it**

ipv36.unipv.it → 193.204.35.27 [AS137 GARR]

ipv512.unipv.it → 131.175.192.3 [AS137/AS198949 RADWARE]

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How to register a (2nd level) domain

Domain registration means adding a node at any level of the DNS tree by purchasing a domain name

Several steps:

1. Choice of the TLD
2. Choice of the Registrar
3. Choice of the name for the new domain
4. Choice of at least two Name Servers (i.e., one primary and at least one secondary)

Domain names are assigned on a *first-come first-served* basis and must be unique

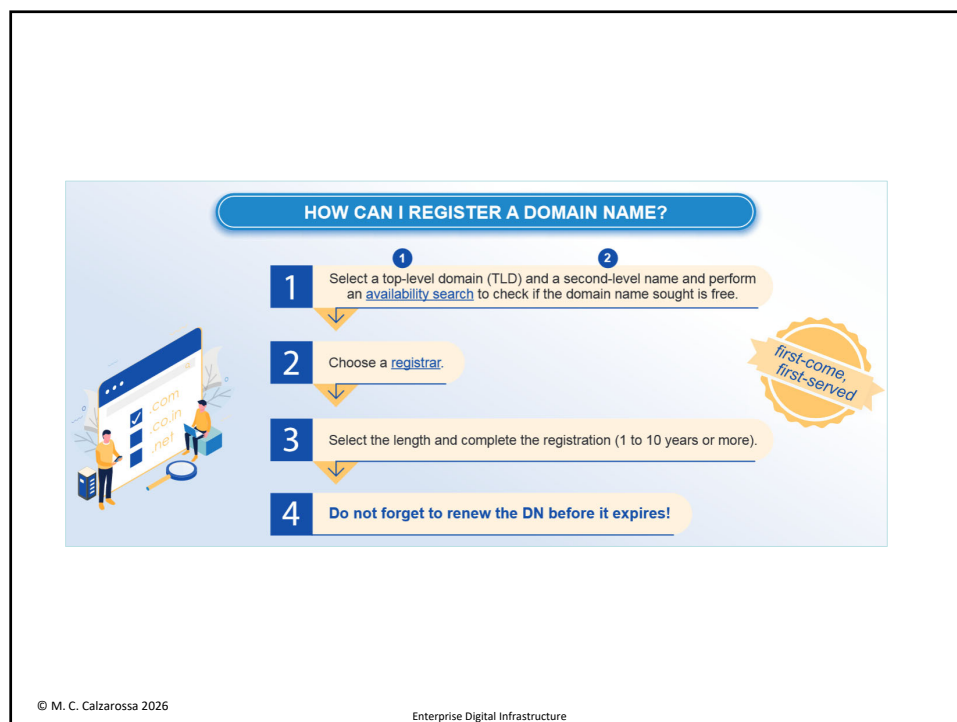
Some registrars offer brokerage services to obtain a domain name already in use

Hostnames inside the domain can be freely chosen (but must be unique)

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34

Rules for registration under **it**

An unlimited numbers of .it domain names can be registered by anyone who is an adult and has citizenship, residence or commercial headquarters in the countries of the European Economic Area (EEA), in the State of the Vatican, in the Republic of San Marino, in Swiss Confederation and in the United Kingdom.

- a .it domain name can consist only of letters from "a" to "z", digits 0 to 9 and the symbol "-" (hyphen) and all the NON ASCII charcters belonging to the charset indicated in the "Technical Guidelines".
- It should not begin and end with the symbol "-" (dash) or start with the character sequence "xn".
- A .it domain name can be composed of a minimum of 3 characters and a maximum of 63.

Not all domain names can be freely registered. Some .it domains are reserved and not assignable:

- names corresponding to regions, provinces and italian municipalities (eg: toscana.it, pisa.it, viareggio.lucca.it, etc).
- names corresponding to services or network resources (eg www.it, internet.it, mail.it, etc.), such domain names can not be assigned even under the "predefined geographical structure".
- Domain names corresponding to "unsponsored" ICANN gTLDs (e.g. com, .net, .info, .name) can only be assigned as second-level domains (e.g. gov.it) for entities that are recognized by the Italian Registry as "competent organs of the state".

Domain names that do not match the official name of a local government body may be registered freely (s.name.toscana.it, ditla.pisa.it).

Domains that identify Italy as a nation can only be assigned to Italian government agencies.

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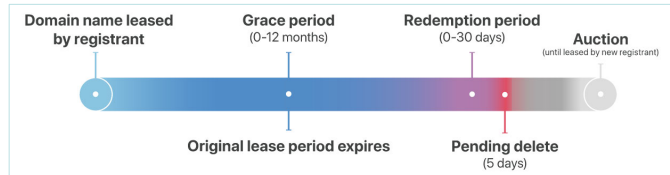
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Registration fee

An annual fee for leasing a domain

Renewal is an important step (e.g., autorenewal)



Watch lists of expired domains (e.g., [expired](#)) & domains for sale (e.g., [auctions](#))

Google Argentina's domain name bought by man for £2

26 April 2021

Google Argentina's domain name was bought by a web designer while the site was out of action for two hours in the country last Wednesday.

Hacking law firms with abandoned domain names

How expired internet domains pose a significant cyber risk to the legal profession

Gabor Szathmari [Follow](#)
Aug 21, 2018 · 20 min read



Expired Domain Allowed Researcher to Hijack Country's TLD

By Eduard Kovacs on January 18, 2021

[Tweet](#)

[RSS](#)

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DNS database

The unique and authentic mappings between hostnames and IP addresses are stored in a wide-area distributed database

Largest distributed database in the world characterized by:

- Scalability
- Decentralized maintenance
- Robustness
- Global scope (i.e., names have the same meaning everywhere)
- No need of atomicity
- No need of strong consistency

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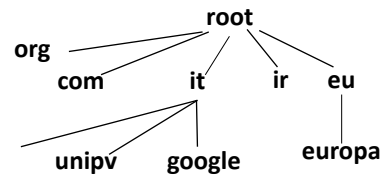
37

Name Servers & database

How are the mappings between hostnames and IP addresses distributed across Name Servers?

Each Name Server stores only a subset of the entire database

Every node of the hierarchy knows about its children (and its parent)



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Who knows what

Root Name Servers know hostnames and IP addresses of **all** Name Servers associated with **all** TLDs

Each Name Server of a TLD knows hostnames and IP addresses of **all** Name Servers associated with its Second Level domains

Each Second Level Name Server knows hostnames and IP addresses of **all** devices registered under its domain name & hostnames/IP addresses of the Name Servers associated with its TLD

Every Name Server knows hostnames and IP addresses of Root Name Servers (4KB of data)

This hierarchical organization allows every Name Server to obtain the IP address (and hostname) of every device

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Content of DB

Information stored in the database are structured in Resource Records

Different types of RRs with a standard format [RFC 1035 “Domain Names – Implementation and Specification”]:

NAME, TYPE, CLASS, TTL, RDLENGTH, RDATA

NAME: label associated with the record (e.g., FQDN)

TYPE: code specifying the type of record [2 bytes]

CLASS: set to a default value **IN**

TTL: how long a record can be stored in cache (e.g., TTL=0 no caching) [4 bytes]

RDATA & RDLENGTH: data describing the resource (as a function of the record type) and length of the data field [2 bytes]

Optional comments: lines starting with “;”

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Mappings

The RR types have to cope with the different mappings between hostnames and IP addresses supported by DNS:

– One-to-one

– Many-to-one

– One-to-many

– *One-to-zero*

virtual hosting

load distribution

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RR types

RR types are subdivided into four categories:

- Basic
- Zone
- Security
- Optional